Whose r^*?

Belief Transmission, Intercept Disagreement, and the Price of Long Bonds

Prospectus, short version · 11 August 2026

Condensed from the full v2 prospectus, which carries the derivations, the qualifications to each test, and the full positioning discussion.

1 The question

The federal funds rate is an overnight rate. The ten-year Treasury yield is not, and its thirty-year decline is usually attributed to forces outside monetary policy. Hillenbrand (2025) shows that the decline accumulated almost entirely inside three-day windows around FOMC announcements, and argues the Fed was revealing its own estimate of the long-run equilibrium rate.

That interpretation makes a structural claim nobody has examined. Writing the policy rule as

$$i_t \;=\; \underbrace{r_t^* + \pi_t^*}_{\text{intercept}} + \phi_\pi(\pi_t - \pi_t^*) + \phi_x x_t + v_t, \tag{1}$$

r^* is the *intercept* of the reaction function, so "the Fed reveals r^*" means the Fed is telling markets about the constant in its own rule. Bauer, Pflueger & Sunderam (2024) estimate the perceived *slope* ϕ_π and show it prices term premia; they absorb the intercept into forecaster fixed effects and say so explicitly. The intercept has been assumed known, assumed common, or differenced away.

> **How does the Fed move beliefs about the intercept of its own reaction function, and what does disagreement about that intercept do to the price of long bonds?**

In three parts: **(1) transmission** — does the published longer-run projection move market beliefs, and in which direction does causality run; **(2) pricing** — does the belief revision reach long yields with the maturity signature of an endpoint shock rather than a stance shock; **(3) disagreement** — is uncertainty about the intercept compensated in the term premium, asymmetrically in the sign of the perceived error?

What the data have already removed

Three results below constrain this severely, and it is better to say so at the front. Fact 3 shows the yield response to be explained is concentrated *before* the published projection exists. Fact 2 shows the signed Fed-minus-market gap has a standard deviation of 10 bp. §6.4 shows neither candidate market-side r^* measure is usable. So Part 1 is estimated where the yield response is indistinguishable from zero and is informative mainly as a null; the dispersion half of Part 3 is Cao, Crump, Eusepi & Moench (SR 934) with a different dispersion series and the signed half has no usable post-2012 variation; and **Part 2 — the maturity signature with its nominal/TIPS split — is the one element that has power on assembled data, can fail, and is not occupied**. The prospectus is organised around it. A second, independent component (§5) requires none of these assumptions.

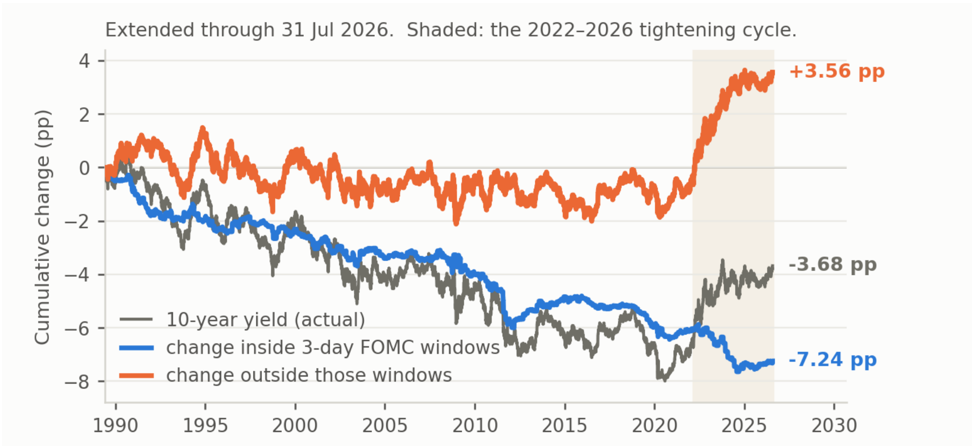

Figure 1: Fact 1. Cumulative change in the ten-year yield from 2 June 1989, split into the part accruing inside three-day FOMC windows and the part accruing on all other days. Shaded: the 2022–2026 tightening cycle.

2 Three facts

Established on assembled data: GSW zero-coupon yields, the ACM daily decomposition, a hand-built FOMC calendar of 300 scheduled meetings and 73 unscheduled actions covering June 1989 – July 2026, the SEP dot-plot archive from 2012, and the NY Fed Survey of Primary Dealers longer-run funds rate from 2013.

2.1 Fact 1: long yields fall only when the Fed is in the room

June 1989 – June 2021: the ten-year fell 6.99 pp, of which 6.16 pp (88%) accrued in three-day FOMC windows representing 11.8% of trading days. Outside, the drift is -0.012 bp/day ($t = -0.18$) over 7,035 days; a randomisation test places the observed figure at the 0th percentile of 5,000 draws.

This is a statement about *conditional means by day type*, not about the permanence of innovations: the two day types have nearly identical daily volatility (6.72 versus 5.60 bp). What distinguishes announcement days is that news arriving on them has carried a systematically negative sign for three decades. Extending to July 2026 constrains any mechanism: over 2022–2026 the Fed raised the target 525 bp and the ten-year rose 2.60 pp, but in FOMC windows it *fell* 1.20 pp — the entire tightening happened on non-FOMC days (Figure 1).

2.2 Fact 2: the two beliefs move together, and asymmetrically

The FOMC publishes its longer-run funds rate projection quarterly; the NY Fed surveys primary dealers for the same object about two weeks before every meeting. Netting the 2% objective from both gives two comparable measures of r^* with no term-structure model anywhere (Figure 2). Across 48 matched meetings: the gap has a standard deviation of 0.105 pp and a range of -0.11 to $+0.30$, exceeding 25 bp at 4% of meetings; dealers move with the Fed nearly one-for-one ($\beta = 0.82$, $t = 6.86$); the reverse relation is present but weaker ($\beta = 0.31$, $t = 3.91$).

The asymmetry is not mechanical — the Desk briefs the FOMC with survey results before each meeting — but it is not a clean lead-lag either. In a horse race the dealer revision loads on both the preceding dot revision (0.60) and the contemporaneous one (0.58), so part of what looks like following is anticipation. Both series are observed eight times a year, and the simultaneity cannot be resolved within them by any market-price measurement.

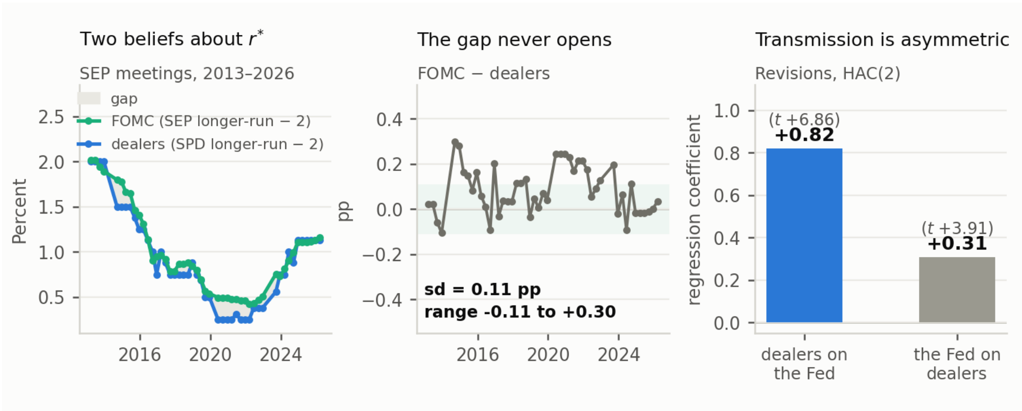

Figure 2: Fact 2. Left: the FOMC's longer-run projection and the primary dealers' longer-run funds rate, each less 2.0. Centre: the difference. Right: revision regressions, HAC(2), 46–47 meetings.

2.3 Fact 3: the phenomenon predates the measure

Period	bp/meeting	t	Cumulative
Jun 1989 – Dec 2011 (pre-SEP)	−2.46	−3.06	**−5.85** pp
Jan 2012 – Mar 2022 (dot falling 1.78 pp)	−0.40	−0.41	−0.23 pp
Mar 2022 – Jul 2026 (dot rising 0.74 pp)	−3.49	−1.53	−1.16 pp

81% of the in-window decline is complete before the dot plot begins. In the decade when the FOMC cut its own longer-run rate by 1.78 pp — when an information channel should be loudest — the in-window contribution is indistinguishable from zero. Every design using the published projection is therefore estimated on a sample in which the motivating phenomenon is absent. That is an argument about the whole of Component A, not one contingency within it.

3 Framework

Define $i_t^* \equiv r_t^* + \pi_t^*$ and $\text{gap}_t \equiv i_t - i_t^*$, the stance of policy. From the definition of a long yield as an average of expected short rates plus a residual, and assuming the endpoints are locally martingales,

$$\Delta y_t^{(n)} = \underbrace{\Delta r_t^* + \Delta \pi_t^*}_{\text{endpoint news}} + \underbrace{\omega(\rho, n)\, \Delta\text{gap}_t + \text{gap}_t\, \Delta\omega}_{\text{stance and persistence news}} + \Delta TP_t^{(n)}, \qquad \omega(\rho, n) = \frac{1-\rho^n}{n(1-\rho)}. \tag{2}$$

Four terms, one observable. Reading window moves as endpoint news requires three separate exclusions. The stance exclusion is defended by noting the pattern survives in the 5y5y forward, where the analogous weight $\omega^f = \rho^{n_1}(1-\rho^{n_2})/[n_2(1-\rho)]$ is small — but how small is an empirical question about ρ:

Half-life of the gap	ω, 2-year	ω, 5-year	ω, 10-year	ω^f, 5y5y
1 year	0.56	0.29	0.15	0.01
2 years	0.73	0.48	0.28	0.09
4 years	0.85	0.67	0.48	0.28

An AR(1) on a crude gap proxy — the one-year yield less HLW r^* plus 2 — gives a half-life of 4.4 years on the full sample and 4.1 post-1995. At that persistence a 100 bp stance revision moves the ten-year about 50 bp and the 5y5y forward about 30 bp, so the far-forward evidence does not by itself separate endpoint news from stance news.

3.1 The maturity signature, and what it can and cannot separate

Endpoint news enters $y^{(n)}$ with a coefficient of one at every maturity; stance news enters with $\omega(\rho, n)$, strictly decreasing in n. So the *shape* of the window response across maturities discriminates, with a slope pinned down by ρ. Three qualifications belong in the statement of the test, not in a robustness appendix.

The martingale assumption is load-bearing. Endpoint news is flat only because r^* and π^* are assumed martingales. If perceived r^* revisions are partly transitory — which learning models allow — endpoint news also enters as $\omega(\rho^*, n)$ with ρ^* near one, and the test discriminates two decay rates rather than flat against declining.

This is close to a factor decomposition. Over 2–10 years, flat is approximately level and declining approximately slope, so the raw test asks what Gürkaynak, Sack & Swanson (2005) and Swanson (2021) have asked for twenty years. The contribution is the *restriction*: $\omega(\rho, n)$ is a one-parameter family, ρ is estimable outside the event sample, and the over-identifying content is whether the ρ fitting the maturity profile matches the ρ fitting the gap dynamics.

The TIPS split is what makes it sharp. $\Delta TP^{(n)}$ has an unrestricted maturity profile, so any shape can in principle be rationalised as a premium response; the test is informative conditional on that response being smooth. Running the same test on the real curve removes $\Delta\pi^*$ entirely, and the nominal-minus-real difference identifies the inflation-expectation component directly. That comparison, not the nominal shape alone, is the element with no close antecedent.

3.2 Two channels, kept distinct

Channel 1, a disagreement premium in σ^2_δ: dispersion about the intercept propagates into the variance of the whole expected path, undamped by horizon, so the predicted term-premium sensitivity rises with maturity and does not decay. This differentiates the project from Caballero & Simsek, whose demand disagreement mean-reverts — and it is also what Cao et al. substantially occupy.

Channel 2, a level effect of the signed gap $\delta \equiv r^{*m} - r^{*F}$: the market evaluates the stance against its own neutral rate while the Fed sets policy against its own. Positive δ means policy looks too easy — expected overshoot, an expected correction, compensation in the inflation dimension; negative δ the reverse. This object is *transitory*, so its yield effect is hump-shaped in maturity, not flat. The defence against Caballero & Simsek applies to Channel 1 only, and that is the channel Cao et al. occupy.

Writing the investor's expected path with the Fed's intercept but his own read of the stance,

$$y_t^{(n)} = \underbrace{r_t^{*F} + \pi_t^*}_{\text{endpoint}} + \underbrace{\omega(\rho, n)\,\mathrm{gap}_t}_{\text{stance}} + \underbrace{\Lambda(n)\,\delta_t}_{\text{perceived error}} + \underbrace{TP^{(n)}(\sigma^2_{\delta,t})}_{\text{disagreement premium}} . \tag{3}$$

Two caveats stated as such. The long-run anchor here is r^{*F}, the *Fed's* estimate, whether or not it is correct — a strong assumption, and the opposite of the accounting argument that the Fed cannot durably move r^*; an anchor $\theta r^{*F} + (1-\theta) r^{*m}$ nests both and makes θ estimable from the maturity profile. And $\Lambda(n)$'s hump shape is currently imposed rather than derived, so as written the hump prediction cannot fail; deriving λ_j from a stated Phillips curve and rule, which pins down the peak location, is scheduled work.

What does come free is the sign split: δ enters expected inflation and the expected real path with opposite signs, so the division of the response between the nominal and TIPS curves identifies the sign channel with no decomposition of the nominal curve.

4 Position in the literature

Paper	**Object**	**What it leaves open**
Hillenbrand (2025, *RFS*)	Window decline is endpoint learning	Never tests the exclusions; no belief data
Gürkaynak, Sack & Swanson (2005); Swanson (2021)	Target/path/LSAP factors from the cross-maturity response	Statistical factors; no restriction tying the loading to an estimated ρ
Bauer, Pflueger & Sunderam (2024, *QJE*)	Perceived *slope* ϕ_π; prices term premia	Absorbs the intercept into forecaster fixed effects, explicitly
Cao, Crump, Eusepi & Moench (SR 934)	Forecaster disagreement about the long-run level; $\sim$1/3 of TP variation	Unsigned; not Fed-vs-market. *Occupies Channel 1*
Caballero & Simsek (2022, *AER*)	Fed-market disagreement $\rightarrow$ sign-dependent policy risk premium	Disagreement is about *demand*, which mean-reverts; no term structure, no TIPS
Cieslak & McMahon; Cieslak, McMahon & Pang	Perceived policy errors raise term premia	Error is about the state and the slope, never the intercept
Savor & Wilson; Ai & Bansal; Lucca & Moench	Announcement-day risk premia in bonds	A *return* premium needing no offset; silent on the sign of the yield change. Central to §5

Unclaimed: disagreement that is Fed-versus-market rather than forecaster-versus-forecaster; a *signed* gap rather than dispersion; and the mapping from sign to *which* risk is compensated. The SEP *longer-run* dot as an asset-price regressor is also effectively unused. But the differentiated space is narrower than that list suggests: the Caballero–Simsek defence applies only to the channel Cao et al. occupy, and Gürkaynak–Sack–Swanson is why §3.1 must be presented as a restricted shape test rather than a new design. Note also that the rule framing is not doing all the work it appears to: every loading in (3) follows from "there is a long-run anchor and agents disagree about it," and since all archived longer-run PCE submissions are exactly 2.0, intercept disagreement reduces empirically to disagreement about the longer-run nominal dot. The framing earns its place by making the differentiation from Bauer–Pflueger–Sunderam precise: same rule, different coefficient.

5 Component A: transmission and pricing

A1. Does the projection move prices? The longer-run dot is published at four meetings a year and not the other four, giving a within-Fed contrast that holds announcement format fixed:

$$\Delta y_t^{(n)} = \alpha + \beta_1 \mathrm{SEP}_t + \beta_2 \left(\mathrm{SEP}_t \times |\Delta \mathrm{dot}_t^{LR}|\right) + \gamma' Z_t + \varepsilon_t, \tag{4}$$

with Z_t containing the Kuttner (2001) surprise, the path factor, and pre-meeting controls in the spirit of Bauer & Swanson (2023). The minimum detectable effect is computed and reported *before* estimation — 57 meetings, roughly 28 with a non-zero revision, residual window standard deviation about 7 bp — so that a null is interpretable. Note that intraday data does not help identify this: the SEP arrives at 2:00 pm in the same tick as the statement and the near-term dots, so intraday windows separate statement-plus-SEP from press conference and nothing finer.

A2. The maturity signature *(core test).* Estimate the window response at every GSW maturity from 1 to 30 years on meetings where the longer-run projection was revised, and test the restriction in §3.1 with ρ first fixed at its gap-process value and then free; the over-identifying test is whether the two agree. Repeat on the TIPS curve and report the nominal-minus-real difference.

Report the estimated profile alongside the first two principal components over the same grid, so the reader can see how much of the result is level-versus-slope and how much is the restriction.

A3. Is intercept disagreement priced? ***(bounded)*** Dispersion varies usefully where the level gap does not: the dealer interquartile range spans 0–0.75 pp and the FOMC's own cross-participant standard deviation 0.22–0.46 pp. Regress long-horizon forward term premia on intercept dispersion, orthogonalised against the Bauer–Lakdawala–Mueller option-implied uncertainty measure and against Cao et al.'s forecaster dispersion; the coefficient of interest is what survives that control. The only element not already claimed is the prediction that the sensitivity does not decay with maturity.

A4. Asymmetry in the sign ***(contingent).*** Fact 2 rules out the simplest version. Three routes, in descending confidence: the pre-2012 Tealbook sample against contemporaneous Blue Chip long-range forecasts, where level disagreement is plausible and the yield response lives; distributional rather than median gaps, since tail disagreement may be larger even when medians coincide; and revision direction rather than level, which with 18 upward revisions gives 21–31% power. If the first two fail, A4 is dropped.

6 Component B: why announcement-day news has one sign

Component A rests on a chain of maintained assumptions. Component B rests on almost none.

An earlier draft framed this as "why do only scheduled-date changes persist?" That is not what Fact 1 establishes. Fact 1 compares conditional means by day type; a driftless random walk has fully permanent innovations and cumulates to zero, so permanence and drift are different objects. The puzzle is **one-sidedness**: why has news arriving on 12% of trading days carried a systematically negative sign for three decades, when those days are no more volatile than any other? That connects directly to Savor & Wilson, Ai & Bansal and Lucca & Moench, who document that bonds earn a disproportionate share of their excess return on announcement days. Their object is a *return* premium, requiring no offset and implying nothing about the sign of the yield-level change; Hillenbrand's is a cumulating *yield-level* effect, coherent only if the yield is non-stationary over the sample. Reconciling them is the contribution, and the 2022–2026 reversal of the outside-window series to +3.56 pp is the first evidence the offset arrives.

Three designs. *Event-time conditional mean path*: the mean daily change at each event time from $t-10$ to $t+10$, with block-bootstrapped bands — not done for US Treasuries, and the shape discriminates build-then-resolve from drop-then-rebuild. *Variance ratio by day type*: a distinct object with a distinct target, adjudicating Hanson, Lucca & Wright against Hillenbrand; it does not test Fact 1. *Stationarity accounting*: report the offset by sub-period.

The competing explanation is Lou, Pintér, Üslü & Walker (BIS WP 1232), who report the effect concentrates in windows *without* long-dated Treasury issuance — dealer inventory rather than monetary policy. This test is cheap but not decisive: Treasury issues on a fixed monthly refunding cycle, so overlap with the FOMC calendar is mechanical and correlated with data-release timing. Step one is counting how many of the 373 windows contain long-dated issuance and checking whether the split is balanced; if not, the result is a control rather than an adjudication.

7 Completed work

1. **The literature's disagreement about the channel is an event-list artifact.** Pan & Peng's day-($t-1$) estimates reproduce to three coefficients (-0.785 / -0.716 / -0.070 against -0.79 / -0.71 / -0.08), but the term-premium share moves from 91% to 46% on the same day,

same data and same decomposition, purely by changing the event list. The 73 unscheduled actions a scheduled-only filter removes carry −2.5 bp of expectations decline per event against −0.07 for scheduled meetings. Neither paper discusses the event list.

2. **The standard decomposition cannot arbitrate.** $\Delta rn_t^{(n)} = c_n' \Delta y_t^{\mathcal{N}}$ with $R^2 = 1.00000$ over 9,282 daily observations, so $\beta^{rn} = c'\beta^{\mathcal{N}}$ exactly and the decomposition adds no event-specific information; $c'\mathbf{1} = 0.529$, so a permanent credible parallel shift is reported as 53/47. Before write-up this needs the Kim–Wright figure and an explicit scope limitation to *spanned* models.
3. **The effect and the measurement do not overlap in time** — Fact 3.
4. **Neither candidate market-side r^* measure is usable.** The ACM 9y1y forward loads 0.287 on the one-year yield ($R^2 = 0.83$) — circular with yields; the survey measure has dealers loading 0.82 on the Fed's own revision — circular with the Fed. This closes the wedge design rather than leaving it open.

8 Risks, timeline, requests

What would kill each component. *Power and the Tealbook dependency*: Tealbook is not a contingency for A4, it is the only sample in which any projection-based design is estimated on the period containing the phenomenon; without archive access Component A reduces to A2 on a weak sample plus a null, and Component B becomes the dissertation. *The maturity test may not separate*: if the endpoint is not a martingale the test discriminates two decay rates, and 28 revisions may not distinguish them. *Simultaneity is not resolved by intraday data*: Fact 2 may remain a descriptive asymmetry. *Positioning*: if Caballero & Simsek's empirics already use the longer-run dot — unverified — the contribution narrows to the maturity signature and the TIPS asymmetry.

Period	Component	Task
Weeks 1–4	—	Draft the reconciliation note (§6.1–§6.2), after adding the Kim–Wright figure and the unspanned caveat
Weeks 2–8	B	Issuance-overlap count, then the event-time conditional mean path with bootstrapped bands
Weeks 4–6	A	Derive λ_j; compute and report the A1 minimum detectable effect
Weeks 6–14	A2	Maturity-signature test on nominal and TIPS curves, with the PC benchmark
Term 2	B, A3	Variance ratio and stationarity accounting; dispersion and term premia
Contingent	A4	Tealbook and Blue Chip, subject to access

Requests. *(i)* Whether the reconciliation goes out as a standalone short piece or becomes a methodological section, and whether the two scope caveats suffice. *(ii)* Tealbook and Blue Chip access — these bind on the whole of Component A, not A4 alone, and this is the highest-value resource request here. *(iii)* Structure: my inclination is to lead with the note, make Component B the first chapter respecified around one-sidedness, and fold A2 into it as the section asking whether the one-sided announcement news is endpoint news — one paper with a fact, a test and a mechanism rather than two half-chapters, with A1, A3 and A4 a second chapter conditional on Tealbook. Does that concede too much of Component A, given it is the part with the genuinely open position?

Key references. Bauer, Pflueger & Sunderam (2024), *QJE*. Bauer & Rudebusch (2020), *AER*. Caballero & Simsek (2022), *AER*. Cao, Crump, Eusepi & Moench, FRBNY SR 934. Gürkaynak, Sack & Swanson (2005). Hanson, Lucca & Wright (2021), *QJE*. Hillenbrand (2025), *RFS*. Joslin, Singleton & Zhu (2011), *RFS*. Lou, Pintér, Üslü & Walker, BIS WP 1232. Lucca & Moench (2015), *JF*. Pan & Peng, SSRN 4764451. Swanson (2021), *JME*. Full list in the v2 prospectus.